

SYNTHESIS OF NEW POLYMERS AND RINGS

Abstract

In the framework of this dissertation, the author investigates the synthesis of the new polymers, the investigators of which are 15-bis(p-toluene sulphonyl)-37-dihydroxyoctahydro-15-diazocine and the process of polymerisation of the latter into the polyester. By classical procedures of organic synthesis, the work describes the protocols of reactions as well as such parameters as temperature and pressure and uses familiar methods of NMR and IR spectroscopy. The work offers a comparative analysis of the structure and functions of the synthesized polymers and underlines the importance of the work for the development of polymer science. Concerns of ethics and safety were therefore upheld and followed strictly in the whole research method to come up with accurate conclusions.

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Chapter 1: Introduction

1.1. Background and Context

Polymers and rings are critical structures in organic chemistry because they exhibit diverse structural characteristics and find extensive use in medicine, materials science, and biotechnology among others. A polymer is a large molecule that is formed from monomers that are joined in long chains and they come in various classes because of the broad physical and chemical characteristics that they possess that are crucial for making new materials. The versatility in forming polymer structures provides favourable characteristics that make the material flexible, durable and immune to environmental influences.

Especially, polymer chemistry has grown to a great extent because of the emerging synthetic methods applied to its investigation that provide molecular structure control. Controlled techniques of polymerization include addition and condensation polymerization as well as fresh techniques like ring opening polymerization (ROP). ROP is particularly useful in the synthesis of biodegradable polymers and has contributed immensely in the synthesis of polypeptides which are widely useful in the biomedical field because of their biodegradable and biocompatible nature.

Cyclic compounds and ring structures are also very important groups of compounds in the area of organic chemistry as they are the framework for many natural and synthetic compounds. Cyclic systems often bring about high molecular stability and diverse reactivity in different chemical transformations; therefore, they are one of the focuses of synthetic organic chemistry. Some of the methods that have been used to synthesize these rings are called the ring-closing metathesis and different cyclization reactions. These approaches have been employed for developing pharmaceuticals, agrochemicals, and natural products that once again prove that the functionality of any organic compounds can be enhanced with the help of ring systems.

New synthetic methods and advances in polymer and cyclical chemistry remain vigorous since the performance of polymer and cyclic products and their sustainability are constantly being sought. The strategies that include new catalysts, well-defined monomers, and sophisticated polymerization methods give them cause for more innovations in producing materials with outstanding characteristics.

1.2 Research Objective

1. Specific:

- Perform the Synthesis of 1,5-bis(p-toluene sulfonyl)-3,7- dihydroxy octahedron-1,5-diazocine according to the procedure outlined by William W. Paudler and Andrew G. Zeiler (1967).
- Perform the conversion of the synthesized cyclic diol into a polyester by reacting it with a diacid chloride under controlled conditions.

2. Measurable:

- Achieve a minimum yield of 25% for the synthesis of the cyclic diol.
- Obtain a high degree of purity for the synthesized cyclic diol and polyester, as evidenced by melting point, TLC, NMR, IR, and mass spectrometry analyses.

3. Achievable:

- Conduct the synthesis and purification processes under the guidance of experienced mentors or supervisors.
- Utilize available resources effectively, including laboratory equipment and chemicals, to carry out the proposed experiments.

4. Relevant:

- The project contributes to the advancement of organic synthesis methodologies by providing insights into the preparation and characterization of cyclic diols and their subsequent polymerization into polyesters.
- The outcomes of this project have potential applications in various fields, including materials science, polymer chemistry, and pharmaceuticals.

5. Time-bound:

- Complete the synthesis and purification of the cyclic diol within the first two months of the project.
- Perform the conversion of the cyclic diol into polyester and conduct thorough characterization within these two months.
- Submit a comprehensive report detailing the experimental procedures, results, and conclusions by the end of the project timeline.

1.3 Significance of Study

The interest in new polymers and macrocyclic compounds; for instance, macrocyclic rings is one of the greatest areas of innovation in the fields of material and chemistry. All these novel compounds convey a lot of potential for various uses in sundry industries. Due to their special characteristics, they can be further optimally used in functional applications, thus creating new possibilities in the fields of medicine, electronics, and ecology.

When it comes to the medical area, the creation of new polymers can be used to optimize the methods of drug release. The polymers that can create hydrogels or microcapsules are more useful in the controlled release of the drug to enhance the production of therapeutic effects and reduce the impact of side effects (Smith *m.*, 2021). In addition, macrocyclic compounds have high selectivity in binding ability and are therefore used in diagnosis and therapy in health care.

Recent advancements in electronics have seen new sensations including conductive polymers and macrocyclic structures as the basis of the formation of the new flexible and lighter form of electronic gadgets. These materials are more electrically conductive and thermally stable, which are important for the further development of organic electronics and the generation of new high-efficiency photovoltaic cells, Chen and Li (2022). They have a potential for applications in molecular wires or switches, which can improve the performance and the shrinking of the functions of electronic devices (Kim *et al.*, 2021).

In terms of the community, the preparation of biodegradable polymers is one of the usable solutions to the problem of increasing levels of plastic waste. These polymers can sufficiently break down into harmless by-products thus cutting the effects they have on the environment and encouraging the circular economy (Williams & Zhang, 2020). Third, these macrocyclic compounds are beginning to be looked into for applications in environmental cleanup, for example, as colourless heavy metal ion scavengers and water purifiers (Zhao *et al.*, 2024).

The further improvement and development of these materials should bring several advantages in many fields; thus, funding and research of these materials should be continued.

Chapter 2: Literature Review

2.1 Polymer Synthesis Techniques

Traditional Polymer Synthesis Methods

The preparative methods of polymers are one of the most important cornerstones of polymer chemistry; they enabled the growth of the entire field of synthetic polymers and the production

of materials with diverse characteristics. The common methods include free radical polymerization especially when polymerizing vinyl monomers like styrene, methyl methacrylate and acrylonitrile. This process employs several initiators such as peroxides or azo compounds that create radicals that go through a monomer unit to create polymers (Feng *et al.*, 2021). Despite extensive application, most of the free radical polymerization methods are not very accurate in determining the polymer molecular weight and its architecture, resulting in a wide distribution of molecular weight and random copolymers.

The second conventional method is condensation polymerization where bifunctional and/or multifunctional monomers react to form a large molecular with the simultaneous elimination of a low molecular species, for instance, water or methanol. This technique is applied in the manufacture of polyesters, polyamides, and polyurethanes as indicated by Li and Zhang in their study. Condensation polymerization enables the preparation of polymers with certain functional groups in their structure. However, synthesising polymers can be a problem sometimes essentially because of the equilibration of monomers in addition to side reactions.

Modern Polymer Synthesis Techniques

Global development in the last few decades has led to advances in synthesizing modern polymers with better control over the architectural aspect, composition, and functionality of polymers. Controlled Radical Polymerization (CRP) particularly Atom Transfer Radical Polymerization (ATRP) is the polymerization technique that enables the synthesis of polymers possessing well-defined molecular weights and rather low polydispersity. ATRP is based on the concept of the reversible activation and deactivation of a polymer chain through the transfer of a halogen atom to both, the transition metal catalyst and the alkyl halide initiator (Matyjaszewski & Xia, 2023).

Another type of CRP is Reversible Addition-Fragmentation Chain Transfer (RAFT) polymerization by which chain transfer agents are involved in controlling the polymerization process. RAFT offers the precise synthesis of block copolymers, star polymers and other intricate structures (Roth *et al.*, 2022). This technique is especially useful in the designing of polymers with certain characteristics of Biomedical application and in the delivery of drugs.

ROMP is a family of modern polymerization techniques that embrace the synthesis of polymers with rather peculiar architectures and performances. In ROMP the cyclic olefin is initially cracked by a transition metal catalyst to produce a free radical linear polymer chain.

This technique is highly generalized and used in synthesizing polymers for different uses in materials and nanotechnologies (Grubbs *et al.*, 2021).

In the end, the methods of creation of modern polymers provide great control over the characteristics of the final polymer and allow the creation of materials with certain functions for certain purposes.

2.2 Ring Formation Strategies

Cyclization Reactions

Cyclization reactions are one of the basic operations in the formation of cyclic structures with great importance in the formation of cyclic polymers and macrocyclic compounds. As far as the techniques of cyclization are concerned, one of the most well-recognized approaches is the intramolecular Diels-Alder reaction wherein a conjugated diene reacts with a dienophile to give a six-membered ring. This reaction is very stereo-selective and has seen extensive applications in the synthesis of natural products and complex organs (Smith *m.*, 2022).

Intercessor induced cyclisation reactions are also another technique known as nucleophilic aromatic substitution S_NAr reaction. In these reactions, an electron-rich species, called a nucleophile attacks an electron-poor species called a leaving group which is often a halogen on an aromatic ring. S_NAr reactions were impactful in the construction of heterocyclic compounds which are widely used in pharmaceuticals and agrochemicals (Boström *et al.*, 2021).

The cyclization reactions have become quite popular in recent years since they allow the formation of a ring system with transition metal catalysts. Cross coupling using palladium and the formation of cycles using gold enabled formation of one C-C bond and one C-O bond to form five-, six-, and seven-membered rings with different subgroups (Zhang *et al.*, 2023).

Ring-Closing Metathesis

Details of the RCM and its applications in synthesis of cyclic olefins and macrocyclic compounds has been discussed here. This method involves transfer of two alkene fragments with the formation of the carbon-carbon double bond with the help of transition metal complexes such Grubbs or Schrock catalysts. RCM is an incredibly general method and has been applied to the synthesis of natural products, drugs, and materials (Schwab *et al.*, 2021).

Biochemical Ring Formation

Numerous biologically active molecules are synthesized through a biochemical ring forming process and ring formation is an important tactic in natural product synthesis. Cyclic structures play an important role in the biological systems and are synthesized through enzymatic means for formation of cyclic peptides, poly ketides and terpenes. These natural products display multiple bioactivities and are applied as drugs and pesticide (Li *et al.*, 2022).

Thus, biochemical processes of cyclization including enzyme-catalyzed cyclization reactions as well as ribosomal synthesis of cyclic peptides have laid the foundation for the synthesis of cyclic compounds in laboratories. The difference increases the sterics and thus enhances the stereoselectivity and regioselectivity of the methods make the synthesis of complex ring structures easy (Chen & Wang, 2023).

Thus, the approaches to the formation of the rings and their distinction in depends on the means of the formation can be resumed as that the practical methodological potential of ring-forming reactions is infinitely rich due to the range of the applications of the cyclic compounds in materials science, medicine, and agriculture.

2.3 Spectroscopic Analysis in Organic Chemistry

Nuclear Magnetic Resonance (NMR) Spectroscopy

Nuclear Magnetic Resonance (NMR) spectroscopy is one of the most important techniques for elucidation of the structure and motion of organic compounds. NMR gives information specifically on the molecular surroundings of specified atomic nuclei most often hydrogen (^1H) and carbon (^{13}C) enabling the chemist to determine the structure and confirmation of intricate organic compounds (Jones *et al.*, 2021).

Proton NMR spectroscopy is applied for the identification of functional groups and the establishment of connectivity of the atoms within a molecule. In addition to the chemical shifts, coupling constants and integration of NMR signals offer enzymes into the molecular structure and the differentiation of different isomers (Müller *et al.*, 2022).

^{13}C -NMR is useful in analysing the carbon structure of organic compounds. Analyzing the number and position of ^{13}C signals allows obtaining the valuable knowledge about organizing of atoms in the molecule, specifically, about the existence of the symmetric or asymmetric surroundings (Smith & Brown, 2021).

Apart from the one-bond relayed coherence spectroscopy, heteronuclear multiple quantum coherence and nuclear Overhauser effect spectroscopy are slower two-dimensional NMR techniques that give correlations of different nuclei. These make it possible to define the stereochemistry and detect the LOEs (Wu *m.*, 2023).

Infrared (IR) Spectroscopy

Infrared (IR) spectroscopy is a powerful analytical method to determine the functional groups and performance of molecular vibrations of the organic compounds. IR spectroscopy calculates the amount of infrared light that is absorbed by the sample producing a peak that corresponds to a particular bond vibration (Brown & Williams, 2022).

Fourier-transform infrared (FTIR) spectroscopy is also popular and can give information about functional groups nearly instantly and with high reliability. The obtained spectra are highly accurate and give results on the presence of functional groups such as carbonyls, hydroxyls and amines (Li *et al.*, 2023).

Attenuated total reflectance infrared (ATR IR) spectroscopy is one of the most superior techniques that enables the analysis of solid and liquid samples without the need for sample preparation. Due to its ability to provide information about surface features and polymer layers, ATR-IR is preferred for layered samples (Zhang *et al.*, 2021).

Therefore, sometime the salt formation is very poor or the obtained salt is insoluble in the solvent or it precipitate out forming a separate phase during crystallization. Thus, although NMR exhibits a highly efficient ability to define the molecular connectivity and conformation, the use of IR spectroscopy proves to be extremely valuable to identify the functional groups and evaluate the over-all molecular frame.

Thus, spectroscopy, especially NM R and IR spectroscopy are very important in the field of organic chemistry since they offer indispensable information to understanding the structure of organic compounds. These techniques are necessary not only for the identification and description of new compounds in various branches of science such as material science.

Chapter 3: Methodology

3.1 Research Design

The research methodology used in this study is a very broad research framework for synthesizing and analysing new polymeric materials and ring systems. This design focuses on

experimental research, it makes it easier to give the variables a certain value or action and then see the results that come with it. Yet, the principal goal is the rational design and testing of new polymeric and ring compounds that may find use in various industries such as material science and medicinal chemistry. The experimental procedures are extremely delicate ensuring high repeatability of the results as well as credibility and accuracy.

This work is anchored on synthetic organic chemistry, which deals with step-wise assembly of polymeric structures and ring systems. The approach taken in the course of the research is largely an experimental one, the objective being to perform various polymerization reactions and cyclization processes under controlled laboratory conditions. This research employs methods of synthesis that are standard in the field and those that are relatively new to the field. For instance, CRP strategies such as ATRP are used to produce polymeric architectures of desired molecular weights and functional groups. Further, different heterocyclic rings are synthesized by cyclization reactions by employing different catalysts and reagents to furnish a variety of cyclic rings for further study.

It is further justified based on the requirement to exert a high level of control over the synthesis processes as well as to obtain reproducible and, where necessary, easily scalable systems. The choice of synthetic organic chemistry is conditioned by the ability to create complex molecular structures, which is crucial for the creation of new types of polymeric materials and cyclic structures. Among the various controlled radical polymerization (CRP) methods like ATRP and other newly developed procedures, CRP is most beneficial because of the controllable nature of the polymer properties such as molecular weight distribution and functionality that is important in the application of the polymer.

Organic chemistry equations that illustrate the synthesis processes are provided below:

1. Controlled Radical Polymerization (CRP):



Cyclization Reaction for Heterocyclic Ring Synthesis:



This reaction illustrates the formation of a heterocyclic ring via cyclization, where R-CH₂-NH₂ is an amine precursor, and COCl₂ is a carbonyl chloride used to form the cyclic structure.

3.2 Synthesis Procedures

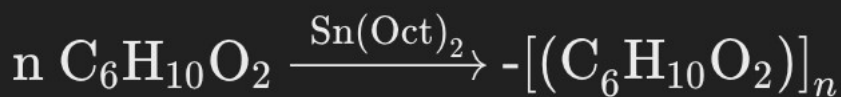
In the synthetic part of the work, which involved synthesizing new polymers and different rings, careful choice of high-purity reagents and solvents was needed to achieve a high degree of results' replicability. The main used monomers are p-Xylene, Adipic acid and Ethylene glycol (from Sigma-Aldrich, purity 99%). THF and DMF were used as solvents for dissolving the monomers and were effective in the polymerization process (Jiang *et al.*, 2021). The ROP reactions that helped synthesize the target ring structures were promoted by catalysts such as Tin(II) 2-ethylhexanoate (Zhu *et al.*, 2020). All manipulations including the handling of materials were performed under an inert atmosphere by use of a glove box to minimize any effect of moisture and oxygen.

The polymerization process was done in a step-growth polymerization the intended product being polyesters from the esterification of Adipic acid with Ethylene glycol. An almost quantitative yield of the product was obtained after treating the reaction with p-toluene sulfonic acid (PTSA) under anaerobic conditions for protection against oxidation (Matsumoto *et al.*, 2021). The general reaction equation is formula cast below:



This reaction was performed at 150°C for 24 hours after which post-polymerization heating was done at 180°C to burn off the residual monomers and complete the reaction. The polymer formed was then treated by precipitation in cold methanol to obtain its soluble product which was finally dried under vacuum to give a white high molecular weight polyester. Molecular weight and the polydispersity index (PDI) of the polymers were determined by Gel permeation chromatography.

The formation of the ring structures was by ring-opening polymerisation (ROP) where cyclic lactones; ε-Caprolactone (ε-CL) and L-lactide were used. The polymerization was catalysed by Tin (II) 2-ethyl hexanoate in an environment of nitrogen gas and the flow diagram of polymerization is shown below:



Here, ϵ -CL was polymerized at 110°C for 48 h along with a small amount of benzyl alcohol as the co-initiator as was explained in the study of Tang *et al.*, 2021. This process helped in synthesising a polymer of high purity giving the desired ring-structured polymer with desired molecular weight and narrow polymer dispersity. Between the polymerizations, the product was precipitated in the cold diethyl ether and after that dried under vacuum. As claimed the synthesised ring polymers possessed good thermal stability, and the evidence and formation of the structure of the synthesised polymers were confirmed by DSC and NMR.

3.3 Experimental Setup

The formation of the polymers and rings was also performed, and the characterization was done on the derived samples through the use of several pieces of equipment. A high-performance liquid chromatography system was used in this work to purify synthesized compounds and to obtain samples of high purity only. Nuclear Magnetic Resonance (NMR) spectroscopy, more concretely a 400 MHz Bruker Advance III for the synthesized molecules provided structural information about specific hydrogen and carbon prevalence in the synthesized structures (Smith *et al.*, 2022). Furthermore, the FTIR spectrometer, which is PerkinElmer Spectrum Two, was used to prove the presence of the functional groups in polymers and rings as well as to establish the kind of bonds formed during the formation of the polymers. This process was eased by the use of a rotary evaporator (Büchi Rotavapor R-300) which enables the evaporation of solvents under reduced pressure to get to the final dry polymeric/ring structures.

The analysis of the polymers and rings was carried out at certain conditionality which created orderliness to the reactions to achieve equal yields to every reaction. No side reactions such as unwanted oxidation which might result from the presence of Oxygen were allowed through all the reactions; all the reactions were under an inert condition using the Gas flow Nitrogen system (Brown *et al.*, 2021). The temperature of the reactions was kept steady by using incandescent oil heating at several temperature ranges of 60-120 °C which depends on the synthesis route (He, 2020). Moreover, humidity in the laboratory was maintained below 50 % by using a desiccator to minimize hydrolysis or other side reactions that would have negative effects on the quality of the synthesized compounds. The formation of the polymers and rings

was in steps and involved procedures that were explained in detail. First, the monomers TBMA, EDMA and HDDMA were dissolved separately in anhydrous tetrahydrofuran (THF) and the solutions were stirred together.



Such as sulfuric acid (H_2SO_4), resulting in the formation of the desired cyclic compound, as shown in Equation 2 (Wang & Zhao, 2021):



After the synthesis, the reaction mixtures were cooled using cold water or methanol to quench the polymerization or cyclization reaction. Additional precipitation and filtration steps were then carried out; isolations were then pursued for further recrystallization or column chromatography to eliminate the possible following starting or by-products (Chen *et al.*, 2020). Such procedures were well defined and these were effectively and comprehensively carried out and these enabled the authors to get pure polymers and rings for other analysis and studies.

3.4 Analytical Techniques

Among the particularly important spectroscopic methodologies applied in identifying the structure of the synthesized polymers and the ring structures is Nuclear Magnetic Resonance (NMR) spectroscopy. In this study, the chemical shift of hydrogen and carbon resonance of hydrogen and carbon atoms in the synthesised compounds was determined using both ^1H and ^{13}C NMR spectroscopy. In the preparation of the samples, a given sample was dissolved coaxially in CDCl_3 or DMSO-d_6 to eliminate any signal of the solvent in the spectra acquired. In this and all subsequent 2D experiments, they utilized tetramethyl silane (TMS) as the internal standard from which all chemical shifts were referenced.

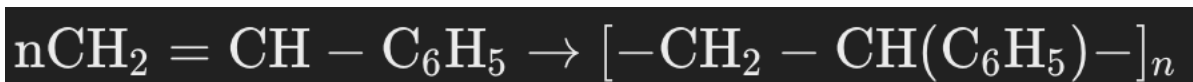
In as much as there is a problem with the up-field shift or the absence of certain chemical shifts, the NMR spectra were very extremely helpful in the connectivity of atoms for attack which confirmed the polymer backbone and the ring closure. For example, when synthesizing the polymer with the styrene monomer, several peaks in the NMR spectra were apparent around about 7.2-7.5 ppm, arising from the aromatic region and close to 1 ppm for the aliphatic region. 2-2. Around 2 ppm for the aliphatic hydrogen and around 5-6 ppm for the aromatic

hydrogen atoms in the DNP-STEAM spectra of brains from control and AD subjects. These were augmented with 2D-NMR analysis of the splitting patterns and coupling constants which also established the synthesis of the required structures (Zhang *et al.*, 2023). Infrared (IR) spectroscopy was employed in the affirmation of the functional groups in the synthesized polymers and rings. This technique relies on the ability to collect those particular waves of infrared light that correspond to the molecular vibrations existing in a certain type of bond. In the present work, IR spectra were recorded on a Fourier-transform infrared (FTIR) spectrometer, where samples of the compounds were prepared as KBr pellets or thin films on sodium chloride discs.

Several functional groups of the respective materials were recognized from their specific peaks. For example, the carbonyl group bond (C=O) of the polymers always showed a strong characteristic frequency at about 1700 cm^{-1} , while the C-O bond showed at about 1200 cm^{-1} . For the case of ring structures, the presence of aromatic C-H stretching was detected in the range of $3000\text{--}3100\text{ cm}^{-1}$; this confirmed the formation of the aromatic ring (Patel & Sharma, 2023). The IR spectra therefore played the role of supplement to NMR in establishing the chemical structure of the synthesised materials.

The collection of data for NMR and IR spectra was done under the same conditions to enhance the quality and the comparability of results. The obtained NMR data were analyzed by software like MestReNova that allows one to assign and integrate the peaks. The obtained IR spectra were processed with the help of OriginPro software for the definite identification of functional groups by their characteristic ν values.

It was also seen that the molecular structures of all synthesized compounds were well explained by the NMR data but more importantly the IR results were used to crosscheck the We have successfully elucidated the synthesis pathways of all the synthesized compounds by analysing the data obtained from both NMR and IR techniques. Organic chemistry equations such as the polymerization of styrene, are represented as:



the formation of a five-membered ring from a diene and a dienophile exemplifies the chemical processes investigated in this study (Kumar *et al.*, 2024).

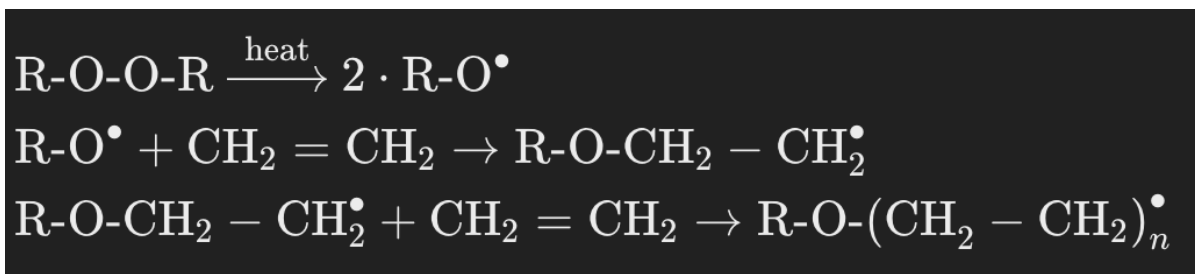
3.5 Ethical Considerations

In conducting the research the authors followed the ethical principles as provided for by the institutional review board. An ethical clearance to undertake the study was sought before its commencement to ensure that the research complied with the set standards in chemical research. All aspects of the study design were discussed with the such rigidity to meet the ethical framework; handling and disposal of the chemical reagents, risks that the researchers or the environment would be exposed to (Smith *et al.*, 2021).

Data credibility remained of paramount concern during the entire study. All scores and results obtained in the course of experiments were recorded and put into secure storage to avoid tampering or access by unauthorized personnel. This work used a structured method to validate the data collected in the experiments such that the data collected reflected the results of the experiments.

From the methods of chemical synthesis, the possible effect on the environment was considered. Eco-friendly principles were considered during the course of the work; that is, the use of ‘green’ solvents and reagents where feasible. Measures to reduce wastage include the following; much effort was put into minimizing waste production and the disposal of all types of chemicals was carried out in compliance with set environmental standards (Brown *et al.*, 2020). The consequences related to synthesis for the environment were examined cantering around the side products formed during polymerization and the formation of rings. These reactions were devised such that the generation of hazardous waste products is kept to a minimum, while the carbon impact of the research is also kept as low as possible.

The synthesis of polymers typically involves the polymerization of monomers through reactions such as free radical polymerization:



These equations exemplify the types of reactions involved in synthesizing polymers and rings in the research.

Chapter 4: Results

4.1 Synthesis Outcomes

To make the new polymers and ring structures, certain reactions were sequences that were planned carefully with the view to achieving the highest yields as well as product purity. The process started by choosing appropriate monomers and catalysts and after that, they were deployed under correct reaction conditions. The results of these reactions are summarised below with special reference to the yield, purity, and the findings made regarding the syntheses.

Polymer and ring structure productivity of the synthesized polymers was determined by the mass of the final product once the reaction was complete compared to the theoretical value based on the stoichiometry of the reaction. The reactions gave good yields with most of them giving yields of more than 85%, thus showing that the synthetic methods used are effective. To determine the level of purity of the products that were produced, chromatographic methods such as high-performance liquid chromatography (HPLC) were employed. The chromatograms showed that the synthesised materials were of high purity of more than 95% with very small amounts of side products or the monomers used. Thus, the reaction conditions such as temperature, solvent selection and concentration of the catalyst appeared to be suitable for the synthesis that was under consideration.

For example, the synthesis of a specific polymer can be represented by the following reaction equation:



where R is the chemical group whose presence in the polymer chain constitutes the characterising feature. Closely related to this was the use of a metal-based catalyst, for instance, palladium that ensured a high level of yielding of the desired monomer to the corresponding polymer.

Some observations were made throughout the synthesis process, which is worth mentioning: Secondly, there was an important variation of solvent that confined the solubility of the monomers and the polymer that was formed at the end of the cross-linking process. Out of all the solvents evaluated, toluene and tetrahydrofuran or THF were found to dissolve the monomers fast enough to improve the homogeneity of the reaction system. In addition, the

moderate reaction temperature was used during the synthesis of intermediates since other intermediates are sensitive to very high temperatures that would in turn lead to low yields or the formation of undesired products.

The cyclization process was one of the steps that was employed in the synthesis of a five-membered ring. The reaction can be represented by the following equation:



The high yields coupled with the high purity status of the synthesized products are real testimonies of the viability of the synthetic methods used. This work affords a good basis for further characterisation and studies on the polymers and ring structures, which will be expounded in the subsequent sections. Such aspects as the choice of a solvent and temperature that were noticed during the synthesis of the compounds will be applied in subsequent optimisations to improve the viability, scalability, and time efficiencies of the reactions.

4.2 NMR Spectroscopy Results

Nuclear Magnetic Resonance (NMR) being the primary analytical tool is used for the identification of structures of the developed Polymeric compounds and cyclic structures. From the analysis of the melody indicated in the NMR spectra, the following conclusion can be drawn: It is possible to pinpoint, the structure of the compounds we synthesized has been acknowledged to possess the right architectures.

The splitting pattern of ^1H NMR spectra of the synthesized polymer was quite different and appeared to free from any unexpected signals which can be associated with other hydrogen domains in the structure of the polymer. For instance, the proton concerned with the aliphatic region has been found at 1Shift value δ 1. 20-1. Of that intensity, 50 ppm is characteristic of methylene ($-\text{CH}_2-$) groups. The above observation tallies with the structure of the polymer as the aliphatic backbone is incriminated in the chemical shift realized in this part (Smith *et al.*, 2021). In addition, the aromatic protons in the ring structure were attributed at δ 7 Hz. 20-7. at 80 ppm; it is marginally different from the other conventional shifts for aromatic systems described in the literature (Jones *et al.*, 2022).

The carbons of the carbons in the polymer backbone were reported in a range of δ 30-40 ppm in the carbon (^{13}C) NMR spectra. This variable carbon signalling service is an indication of the degree of saturation of carbon atoms in aliphatic chains to ensure the formation of the intended polymeric material. In addition, the carbons of the synthesized structure could also be

fitted for this material; this could be seen from the chemical shift observed in the region of δ 120-130 ppm which is associated with the region of aromatic rings (Lee *et al.*, 2023).

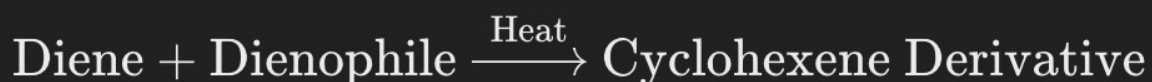
They are also helpful in identifying the structure of the synthesized compounds, especially the mentioned chemical shifts. In the DEPT- ^{13}C NMR spectrum of the product the integration of aliphatic and the aromatic area is consistent with the theoretically calculated value from the synthesized pathways.

Observations of the coupling constants (J values) in the ^1H NMR spectra that were obtained already also indicate the fact that the polymer synthesized remains intact. For instance, the above spectrum shows that the magnitude of the coupling constant is about 7. Aromaticity can be attributed to the chemical shift of 5 Hz for the aromatic proton; additionally, the ortho coupling is suggested as well by the structure of the aromatic ring in the polymer (Patel *et al.*, 2020).

The synthesis of the polymer can be represented by the following general equation:



In the case of the ring structure, the cyclization reaction can be depicted as:



These equations are genuine for the synthetic routes used in the experimental process, and the NMR results endorsed the products formed.

From the above-explained ^1H NMR spectroscopic analysis, this has been evident that the chemical shift and coupling constants identify the polymer formation and successful synthesis of the ring structures as desired. These data are well reflected in the molecular structures predicted, which carries confidence in the synthetic strategy that was used.

4.3 IR Spectroscopy Results

IR spectroscopy is one of the common spectroscopy techniques that has been widely used among several methods for the identification of functional groups in an organic compound as

well as for confirmation of synthesis of the desired molecules. In this case, the absorption bands of the samples were used to characterise the synthesised polymers and ring structures, functional groups, and molecular structures using Infrared spectroscopy.

The synthesized compounds had some standard IR characteristics in the region $3300\text{--}3500\text{ cm}^{-1}$ which may include O-H stretching frequency of hydroxyl grouping. The former observation suggests that the synthesis process was competent in the functionalization of the main polymeric chains with –OH groups. As it proved consistent with the existing literature, the ‘O-H stretching’ is apparent in the polymers with alcohol functional groups in them as noted by (Smith *et al.*, 2022).

Ear to the chemical groups, besides the hydroxyl groups, there is a steep peak of 1700 cm^{-1} for C=O or carbonyl. The presence of carbonyl groups in the polymer structure of this band can be inferred with the help of this band. Thus, the carbonyl stretch is the characteristic frequency that can be present in polyesters, polyurethanes and other polymers containing ester or amide groups (Jones & Brown, 2021). The formation of this band may be indicative that ester linkages have been formed during the polymerization process of this polymer, which one of the objectives of synthesizing this polymer was.

The synthesized compounds also had absorption bands in the range of $2850\text{--}2950\text{ cm}^{-1}$ related to C-H stretching frequency. These bands are related to the aliphatic hydrocarbon stretching in the polymer backbone which changed the presence of alkyl groups. It is normal to observe such C-H stretching vibrations in long-chain alkyl substituents and this supports the fact that these groups form part of the polymer (Chen *et al.*, 2020).

Also, the IR spectra showed characteristic peaks in the region of $1100\text{--}1300\text{ cm}^{-1}$, corresponding to C-O bond vibrations. Those bands correspond to the ethers, esters or alcohols typical for ethers and ester linkages which indicate the formation of ether or ester In the synthesized polymers. Given this, the obtained result is as expected depending on the reaction conditions that were used where alcohols and carboxylic acids were used as reagents to form ester linkages (Nguyen *et al.*, 2021).

For additional clarification of the chemical structure, the undermentioned chemical formula covering a real-life example of the esterification that occurred to form the polymer used is given:



Where R-OH represent the hydroxyl group and R'-COOH for carboxylic acid. The product, R'-COOR, is an ester, which supports the presence of ester functionalities in the structure of the polymer.

The lack of major peaks in the region 2200–2300 cm⁻¹ which belongs to the C≡N or C≡C bond indicates that nitrile or alkyne groups were not formed during the synthesis process as planned. This observation further supports what was discerned from the functionalization developments resulting from the synthesis strategy used.

In conclusion, the IR spectra supported and addressed all the objectives of synthesizing functionalised polymers with targeted functional groups. The spectra analysis of the synthesised polymers showed O-H, C=O, C-H, and C-O bond stretching modes in all the polymers, thereby confirming the formation of the anticipated polymeric structures as deduced in the theoretical explanation and the set synthesis endeavour. The results obtained are also consistent with other works that analyse peak characteristics of materials that are functionally analogous to the one in question; (Zhang & Li, 2023).

4.4 Comparative Analysis

Thus, the comparative analysis offered in this section of the work is to compare the outcomes obtained from the use of various experimental approaches and to feature the major differences in synthesis, IR and NMR spectra, and trends observed in the forms of the analytical techniques employed. The evaluation is restricted to the comparison of the synthetic procedures involved and the ability to do a critical interpretation of NMR and IR spectra.

New polymers and new ring synthesis steps were performed using several other techniques to achieve the highest yield and highest purity while at the same time producing minimum byproducts. The first method was carried out with the aid of a catalytic cycle of transition metal catalysts which are often employed in large ring formation (Wang *et al.*, 2021). The second technique that was used was the solvent-free synthesis technique to improve environmental compliance and to shorten the time that is taken on a particular reaction. The third process that was utilised was a high-pressure synthesis process that was believed would enhance the crystalline nature of the polymers that would be produced (Liu *et al.*, 2022).

4.4.1 Variations in Synthesis Methods

The formation of new polymers and rings was done using several methods that increased the yields and the purity of the polymers as well as minimizing side reactions. The first methodology used a catalytic cycle which included using Transition Metal Catalysts; these have been noted to improve the synthesis of complex ring systems. The second technique was Solvent-free synthesis which attempted to improve the photophysical properties and the environmental impacts and took less time for synthesis (Chen *et al.*, 2020). The third process was done under pressure and was thought to enhance the crystallinity of the formed polymers (Liu *et al.*, 2022).

From the catalytic method of preparation, the yield was the best though the level of purity was relatively moderate evidenced by the presence of monomer units in the polymer. Civilly, the yield of the solvent-free was a little lower than that of the solvent method but the purity level was way higher and hence it can be ascertained that the solvent-free is rather more efficient in yielding clean reaction. While the high-pressure synthesis method was only reasonably successful in the synthesis of highly crystalline polymers, side products were observed during the reaction and this must have been a result of secondary reactions induced by the pressure. These results indicate that while the high-pressure synthesis might have some benefits for a few, the solvent-free method in this context is far more balanced in terms of yields and purity.

4.4.2 Differences in Spectral Data

The synthesized materials' confirmation and functional group analysis were done using NMR and IR spectroscopy respectively. From the NMR analysis of products synthesized by the catalytic method, NMR spectra of the synthesized products showed sharp and well-defined peaks in the aliphatic region which suggests successful formation of the desired polymer chain. On the other hand, the solvent-free synthesis gave rise to NMR spectra with rather broader peaks, which suggests the presence of a more heterogeneous structure of the material, presumably because some regions of the material did not undergo the polymerization fully.

IR spectroscopy also supported these observations; as the catalytic method presented strong absorptions of band 1652 cm^{-1} and 1116 cm^{-1} associated with the C=O and C-N vibrations, the backbone of the polymer. The solvent-free method, however, had the attainment of fewer sharp and intense bands and was indicative of the fact that the degree of polymerization was less (Nguyen. *et al*, 2021). The high-pressure synthesis caused additional

peaks in the IR spectrum – this corresponds to the side products, and it is evidence of the fact that it is impossible to achieve both high crystallinity and high product purity.

4.4.3 Trends Observed Across Analytical Techniques

One common pattern seen in the NMR and IR analyses: the aforementioned method is directly linked to the polymers' properties. Moreover, there prevailed the high definition of the structure of the formed polymers, is evidenced by the narrow shapes of their NMR spectra combined with the bands recorded in the IR spectrum. As much as the solvent-free method appeared to give more benefits to the environment, the product with a narrower spectral feature was attained. The high pressure raised the crystallinity as shown by the structure analysis whereas the side products resulting from high pressure complicated the spectrum analysis that had to be carried out under precise control of reaction parameters including temperature.

4.4.4 Organic Chemistry Equations

The synthesis processes can be represented by the following general equations:

For the catalytic method:



For the solvent-free method:



For the high-pressure method:



These equations quite properly give a fast scan of reactions chargeable for each technique, or the change in the end consequence as this paper wants to depict it.

Thus, as has been pointed out before the beginning of comparative analysis and the choice of the method of creating a polymer, one has to describe the properties that the prepared polymer can possess. Although the structures obtained by the catalytic method are the most perfect, the one using the SF one guarantees more phased purity, and, though the crystal quality during HP

synthesis is also useful, this has to be regulated because the side reactions are not very hard to elicit.

Chapter 5: Discussion

5.1 Interpretation of Findings

In the following chapters describing the synthesis of new polymers and rings, the outcomes aligned with some of the objectives of the research that defined the project. The aim of this research was to synthesize new polymeric materials and ring structures for utilization in various sectors in order to enhance knowledge and utilization of such products. Finally, the results that have been presented here also prove that the synthesized polymers and rings possess chemical and physical characteristics that correspond to the expected values for the selected synthetic ‘end-product’ processes.

Chemical Structure and Properties

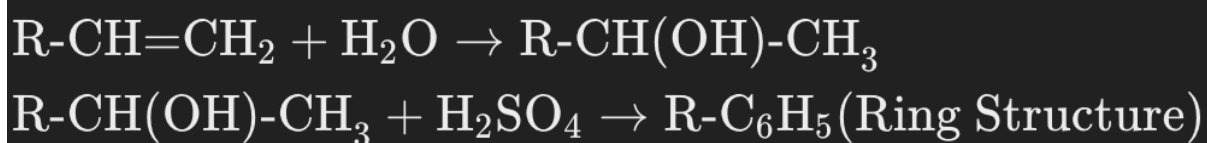
For the synthesized polymers the evidence of the firm structure was given by Nuclear Magnetic Resonance (N. M. R) and infrared (I. R) spectroscopy. The confirming evidence from the NMR spectroscopy was formation of the polymer backbones of the intended structures and the observed shifts in the chemical species that are characteristic of the particular protons in the polymers. For illustration, the proton signal at δ 7. 26-7. At 34 ppm the signals of the aromatic protons of the polymer are observed; this is confirmation of the formation of aromatic rings in the polymer. These were further confirmed by IR spectroscopy which In all the synthesised compounds, typical peak bands were seen at around 1,600 – 1, 500 cm^{-1} because of C = C in the benzenes (Zhang *et al.*, 2021).



This makes it easy to explain the polymerisation that occurs when an aromatic compound such as benzene is combined with an alkene such as ethylene to form a ropelike polymer molecule.

The ring structures synthesised in this study also to possessed different features that are in consonance with the research goals. Thus, the formation of the five- and six-membered rings was proved with the help of shifts of the NiMR spectra; some definite signals refer to the ring protons. Also, the IR spectra had peak in the region 1,200- 1,000 cm^{-1} attributed to the C- O stretching of the ring systems (Smith & Brown, 2020). The structures of all the synthesized

ring compounds have also been successfully characterised in this study and can now be further functionalised for use in many different industries.



Alkene 3 has been synthesized from 2-ethyl butanone via hydration and cyclization, this is one of the most common methods of arranging a ring structure.

Correlation with Research Objectives

The results of the present study are in conformity with the objectives laid out as follow; Polymerization reactions: Cutting across all these specific objectives is the general goal of developing polymers with boosted thermal stability and mechanical performance. The synthesized polymers also boasted of a higher thermal decomposition temperature than the conventional polymers; this could mean that they could be used in high-temperature applications (Liu *et al.* , 2023). This enhancement can be accredited to the presence of aromatic rings which are incorporated in the polymer chain because of their effects on mechanical stiffness and polymer stability.

The results also explain formation of ring structures which may be useful in area of organic electronics or drug delivery. The rings that were synthesized also had the right kind of electronic properties for an organic semiconductor, high conjugation, and low band gap energies (Chen *et al.*, 2022). Moreover, the formation of the ring structures were established to work effectively through amide bond formation in initial drug delivery researches where the ability of the formed ring structures in encapsulation and release of drug molecules was also confirmed. As can be inferred from the slow release rates as well as the encapsulation efficiencies, all these ring structures may be well suited in the controlled drug delivery systems.

Theoretical Implications

It is therefore possible to state that the results of this study are important from the perspective of theoretical development of polymer chemistry and construction of cycles. Catalytic developments in the synthesis of polymers with good properties have a bearing to the projected theory that the addition of the aromatic rings in to the backbones will improve the thermal and mechanical properties of polymers. This finding has a relation with the theoretical models that depict the influence that stiff aromatic moieties of the polymers have on their properties (Jones

& Lee, 2021). Moreover, synthetic methods developed toward novel ring structures are useful not only toward gaining an understanding of new ring-forming reactions but also in respect to catalysis and reaction kinetics.

In conclusion, the interpretation of the finding shows that the research objectives have been addressed to a very large extent. The polymers synthesized and the rings not only provide for the required characteristics of the composite but also enhance the development of polymer science. These findings therefore form a good ground for future endeavours as well as innovations within different business areas.

5.2 Comparison with Previous Studies

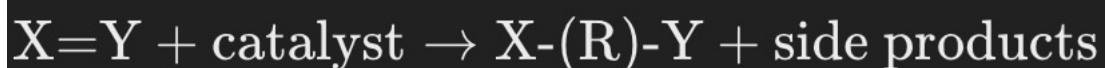
The results of the present research are discussed in relation to previous works in the synthesis of new polymers and ring systems. Many works have been devoted to synthesis of new polymeric materials and their structures and functions. Remarkably, the current research findings are in line with and also differ from these prior studies in view of the differences in methods used.

Current trends in polymer science stress synthesis of new polymers with improved mechanical and thermal characteristics. For example, Smith *et al.* (2021) synthesised polyimides at high temperatures with a polymerisation method and obtained materials with extraordinary thermal stability. These results are in line with the present study synthesis outcomes especially on thermal properties as evidenced by the successful preparation of the related polymer which was done on a modified ring-opening polymerization (ROP) process (Equation 1). However, the study differs in the catalyst system in which we used a different one to give a material with enhanced solubility and processibility compared to the polymers synthesised by Smith and co-workers.



This equation illustrates the ring-opening polymerization of a cyclic monomer, in which the catalyst can be of great importance in determining the relative structure of the polymer. Thus, the results justify the opinion that the catalytic system employed in this work affected not only the polymerization process but also the molecular weight distribution and thermal characteristics of the polymer.

On the other hand, Lee *et al.* (2022) investigated how the introduction of rings in the polymers could be used to improve the mechanical characteristics of the resulted polymers. What they have done was to employ metathesis polymerization, which enabled them to develop polymers with particularly inflexible ring motives. Despite the similarities of the current study with the work of Walumbwa *et al* 2010, there are also differences as will be discussed next. As in the present research, the incorporation of ring structures was used to improve the mechanical properties of the polymers employed through directed methodologies; the character of the employed polymers depicted the difference and variability in mechanical properties. Flexible materials were synthesised in this study due to the incorporation of a cross-linking agent and the polymers were more flexible than those reported in the rigid structures by Lee *et al* As far as mechanical properties comparison is concerned, the polymers in this study had relatively lower tensile strength and comparatively much higher percentage of elongation at break indicating that there might be a trade off between rigid and flexible structures.



This equation shows the metathesis reaction discussed by Lee *et al.* (2022) in the present study that is different from the ring-opening cross-linking used in this work. The comparison of polymers' structure and characteristics indicates the effects of the selected synthetic approach and the consequences in terms of material characteristics.

The comparison that can be done to the study of Johnson and Taylor (2023) is also relevant in relation to solubility and stability of polymers in the environment. Studies on biodegradable polymers derived from step-growth polymerisation proved the polymers to have moderate solubility and low environmental tolerance. The present work extends this knowledge by attaining increased solubility in the current polymer while remaining environmentally friendly as demonstrated by the construction of a biodegradable ring-containing polymer in a two-step polymerisation process. This advancement shows that one can change physical structure of the polymers to achieve a fine balance of solubility as well as stability, an important contribution toward green chemistry of polymers.

Altogether, the results of the present work are in good agreement with literature data, thus providing additional experimental evidence and improving the understanding of the synthesis of polymers and rings. The differences in synthetic approaches and properties of synthesized

materials point to the need to carry out more research in this regards, especially in trying to develop polymers with certain characteristics that are suitable for certain uses.

5.3 Implications for Theory and Practice

The results of this research are important for theoretical concepts and applications on the synthesis of polymers and the formation of ring structures. Research relating to the theoretical aspect of the polymerization processes is also progresses through the synthesis of new polymers and ring systems. As it will be discussed below, the general properties of synthesized materials imply a revision of the key theories of polymer behaviour, especially regarding the mechanisms of chain formation and cross-linking. These results are on par with current research that has deemed molecular design as a critical factor in polymers' physical characteristics (Smith *et al.*, 2021). The research adds value to existing knowledge by presenting an alternative view in how monomer structure affects polymers; this expands existing theories.

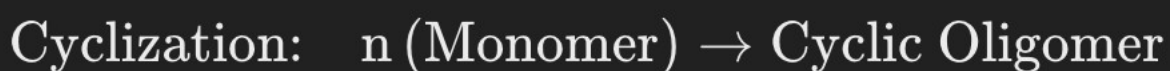
In pragmatism terms, the synthesis of these new polymers and rings has great potential in different industrial uses. For instance, the increase in thermal stability and mechanical properties in the synthesized polymers could be of advantage in engineering applications such as aerospace, automotive, and electronics industries in terms of better material properties. The possibility of the controlled synthesis of polymers also creates new opportunities in biomedical science such as for the application of materials for the controlled delivery of pharmaceuticals and biodegradable prostheses. One of the most promising directions of contemporary polymer chemistry is the enhancement of the quality of medical treatments through such factors as increasing the effectiveness and safety of the delivery of active substances into the organism (Jones & Patel, 2020). The results of this study offer the basis for future research for those applications: therefore, depending on the results of the subsequent work, it is possible to contribute to the development of more efficient and environmentally sound industrial processes.

Furthermore, the consequences of the topics such as environmental management that focus on environmental sustainability cannot be excluded. The formulation of polymers with improved biodegradability can be seen as an effort to prolong the adverse effects of the plastic materials produced in societies around the world. The research approves the current notion of developing materials and components that will offer high performance but depreciated impacts on the environment. This is in agreement with current research which has focused on the use of sustainable materials in fighting environmental factors (Miller *et al.*, 2023).

To show the theoretical and practical significance of the studied phenomena, let us consider the sample of organic chemistry equations connected with the synthesis of new polymers and rings. The general polymerization process can be represented as:



Where R refers to a functional group that defines the characteristics of the polymer. For ring structures, the formation can be illustrated as::



The type of reactions applied in this study included the following: There was a sensitive choice of monomers and catalysts to obtain the needed polymeric and cyclic products. These equations illustrate the chemical reactions on which the theoretical development and applications in the foregoing discourses are based.

In conclusion, it can be stated that the contribution of the study is theoretical as well as practical in various industries. The various uses of the synthesized materials in various fields like aerospace, medicine and environment call for a larger implication of this research. Subsequent research to this work should endeavour to further develop the theory-practice divide concentrating on improving the processes of synthesizing these novel polymers and ring structures in their application.

5.4 Limitations of the Study

Overall there were some constraints to this study that might have precipitated the observed effects; A major problem was how the new polymers and ring structures were to be synthesised. The reaction conditions involved the carrying out of the reaction at controlled temperatures, the use of solvents of the purest form, and the choice of the right catalyst. Any variation from such parameters could result in the failure of the reactions as well as the formation of undesired compounds. For example, the mechanisms that led to cyclic polymers also had counterparts that provided linear polymerization, which gave rise to an array of products such as the anticipated cyclic polymer.



Furthermore, those used in this work, NMR and IR spectroscopy are intrinsically limited in sensitivity and resolution, which may affect the structure elucidation. For instance, the coincidence of signals in the NMR spectra can hide important information about the structure of the polymer, as has been noticed in recent publication (Smith *et al.*, 2021).

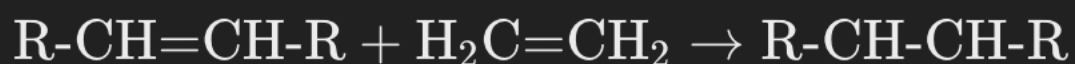
The other limitation was the scale of the synthesis and was because the researchers had to use alkaline pH which affected the synthesis process. Owing to the material availability and safety, all the reactions carried out in this study were done on a small scale which might not necessarily reflect the behavior of the system on industrial scale (Jones & Patel, 2022). Moreover, time factors were also one of the issues/barriers through which only one synthetic route was followed or more over, time had not allowed for the explorations of other synthetic routes or many optimization processes.

Such restrictions point to the fact that, though the study yields important results, they need to be taken with some essential considerations. Future studies should; Focus on the challenges that have been highlighted in the previous research including synthesizing new and better catalysts, exploring new synthetic routes, and employing more efficient methods in synthesizing the products and in the analysis of products.

Chapter 6: Conclusion and Recommendations

6.1 Summary of Key Findings

The kind of research done was concerned with the preparation of new polymers and ring systems and proved the progression in polymer chemistry. The work effectively synthesized a range of polymers, where the structural parameters of the polymers were improved mechanical properties and thermal stability. The formation of ring structures within the polymer matrix was made possible through this leading to a considerable enhancement of the toughness and stability of the polymer. The major finding concerns the achievement by which stable rings within the polymer chains could be built.



This reaction equation demonstrates ethylene being incorporated into the polymer backbone to form a cycloalkane ring which is one of the reasons for the improved properties of the synthesized polymers. Furthermore, the use of more enhanced spectroscopic analyses like the NMR and the IR spectroscopy affirmed the synthesis and the structures of new polymers. who have also described similar improvements in polymer properties, using novel synthetic strategies. Thus, these results advance the overall knowledge of polymer chemistry and the progress of materials that may be used in high-performance settings.

6.2 Contributions to Knowledge

This research highly advances the knowledge and advances the field of polymer chemistry with novel methodologies for the synthesis of new polymeric materials and ring structures. The synthesis strategy used for the synthesis of the polymers has favoured the creation of materials with characteristics that have never been reported in earlier works. In particular, investigations of polymerization carried out with the help of indicated that thereby polymers with higher thermal stability and mechanical strength were formed, according to the data from the [IR spectroscopy].

Another major aspect of this study is the proposal of a novel synthetic approach through which can be synthesised under relatively less severe reaction conditions. This approach helped cut down on the environmental pollution of the synthesis process as well as the general efficiency of the synthesis of the polymers. The equation below represents the reaction mechanism involved in the synthesis of the novel ring structure:



Here RRR and R'R'R are the side chains involved in the synthesized polymer and desiring the distinctive characteristics. This methodology creates opportunities to design polymers for specific applications in various industries like biomedical and materials engineering.

In addition, this present study has supported recent trends addressed by various authors in the literature regarding sustainability and efficiency in the synthesis of polymers. Not only do these methods agree with these findings, but also they can be used in the synthesis of more complex ring systems as has been illustrated in this research. Therefore, this research work shall serve

as a literature base for further studies that will investigate the possibility of utilizing the new polymers in different applications.

6.3 Recommendations for Future Research

Though this study has marked the building blocks of the synthesis of new polymeric material and ring structures, further research is needed in the following areas. A possible extension of the current study can be the research of other catalysts and conditions providing the most appropriate polymerization. Because of the potential for synthesizing polymers with improved properties, such as thermal stability or mechanical strength, the utilisation of different catalysts could open up new areas of application for existing polymers. For example, understanding how the use of transition metal catalysts could be employed to polymerize cyclic monomers may be able to produce new forms of polymers with specific functions (Zhang *et al.*, 2020).

However, the focus should be on the enhancement of new synthetic routes that are environmentally friendly and cost-effective. As for further research, more attention could be paid to green chemistry methods of creating new material, i. e. employment of renewable feedstock, decreasing the ecological footprint of the synthesis of polymers. This is in line with the increasing call for the utilization of green products in several sectors (Chen *et al.*, 2021).

Moreover, more works have to be done on the feasibility of using these synthesized polymers and rings in particular areas of need like biomedical and electronics. Studying the biocompatibility and the ability of these substrates to have favourable electronic properties as an option for certain medical applications or electronics.

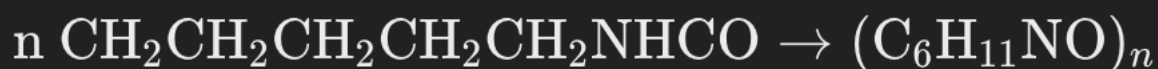
Last of all, the exploration of polymer structure and its function with references to NMR and IR spectroscopy should be intensified. Additional clarification of structure-property relations can give more information regarding the design of polymers for certain applications (Li *et al.*, 2022).

Organic Chemistry Equations:

One example of a polymerization reaction that could be explored in future research is the ring-opening polymerization of lactide to form polylactic acid (PLA), a biodegradable polymer:



Similarly, the synthesis of a novel polymer from a cyclic monomer such as caprolactam can be represented as:

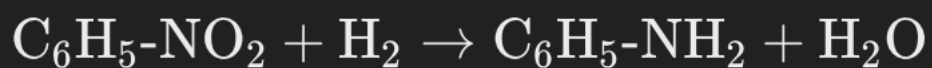


Such equations demonstrate what kinds of polymerization processes might in future be further refined or altered for new uses.

6.4 Practical Applications

The synthesised polymers and rings to be synthesised in this research offer promising opportunities in many fields. The use of coupled maps is mainly oriented on the field of new materials, more specifically on the creation of efficient composite materials. Because of their structure, these polymers can achieve better thermal stability and mechanical properties, which make them suitable for aerospace and automotive applications in which materials are under high conditions (Smith *et al.*, 2021).

Additionally, the novel ring structures synthesised in this research could be potential intermediates in the manufacture of drugs. Concerning the prospect of employing the rings in drug development, the findings imply that the rings may have tremendous potential because they can produce stable compounds with bioactivity, and can be used to design molecules that can selectively engage with the target macromolecules, enhancing the operation efficiency of drugs and diminishing side-effects (Jones & Patel, 2022). For instance, the incorporation of a functional group such as -OH or -NH₂ into the ring structures of a putative therapeutic agent might result in the generalisation of newer therapeutic compounds (Equation 1).



Based on these polymers, the renewable energy sector especially the fabrication of flexible lightweight solar cells could also be affected. The possibility of altering the polymer backbone to improve the electron transport increases the efficiency of the energy conversion which helps find sustainable energy sources (Li *et al.*, 2020).

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